Data Visualization for Research

Presenting Results Effectively

The Whats and Whys of Data Visualization

Introduction

Data Visualization is the **graphical representation of data.** It involves using charts, graphs, maps, and other visual elements to reveal patterns, trends, outliers, and insights more effectively than raw numbers.

There are two different approaches to data visualization:

- **Exploratory:** Your purpose for visualization is the exploration and understanding of trends in your data, either for the purpose of data analysis or the preparation of it.
- **Explanatory:** Your purpose for visualization is to clearly and effectively communicate something about your data to a wider audience.

Why Visualize Data?

01 Simplify Complex Data
It turns large and complicated data into visual formats like charts and graphs, making the information easier to understand.

02 Reveals Patterns and Trends
It helps identify trends, relationships, and patterns that are not easily seen in raw data or tables.

03 Improve Communication
It makes it easier to explain data insights to others, especially those who may not be familiar with the technical details.

Limitations

Oversimplification Risk
Important context or nuances may be lost when data is reduced to visual form.

Bias & Misleading Design
Truncated axes, cherry-picked data, or misleading chart types can distort interpretation.

Cognitive Overload
Complex or cluttered visuals can confuse more than clarify.

Data Quality Dependency
Poor or incomplete data leads to misleading visuals, no matter how well designed.

Tools for Visualization

- **Programming:** R (ggplot2, Shiny), Python (Matplotlib, Seaborn, Plotly, Altair)

- **Web based tools:** Javascript libraries like D3.js

- **Third Party Apps:** Microsoft Excel, Tableau, Power BI

- **Temporal, Geospatial, and Exhibition tools:** Omeka, StoryMap, Timeline.js, WorldMap, and Carto

What makes a Good Visualization

Pre-Attentive Attributes

A human can distinguish differences in length, shape, orientation, distances, and color readily without significant processing effort; these are referred to as **pre-attentive attributes.** Compelling graphics take advantage of the attributes.

There are 4 pre-attentive visual properties: **Form**, **Color**, **Spatial Position** and **Movement**. From these four, we are able to derive specific visual features that the visual system can detect instantly.

Specific Pre-attentive Visual Features in Data Visualization

Quantitative Messages

Quantitative Messages describe the type of insight or comparison a user wants to extract from numerical data, guiding how the data should be visualized.

Each message type is best conveyed through a specific kind of graph or chart, which helps highlight the intended relationship or pattern.

There are **8** types of quantitative messages.

Quantitative Messages

01 **Time-Series**
Shows how a variable changes over time e.g., temperature over 10 years
→ Line chart

02 **Ranking**
Orders categories by size (ascending or descending) e.g., top-performing sales reps
→ Bar chart

03 **Part-to-Whole**
Shows proportions of a whole (percentages) e.g., market share by brand
→ Pie chart, bar chart

04 **Deviation**
Highlights difference from a reference point e.g., budget vs. actual expenses
→ Bar chart

Quantitative Messages

05
Frequency Distribution
Shows how often values occur within defined intervals e.g., stock returns grouped in ranges
→ Histogram, boxplot

06
Correlation
Shows the relationship between two numerical variables e.g., inflation vs. unemployment
→ Scatter plot

07
Nominal Comparison
Compares values across unordered categories e.g., sales by product code
→ Bar chart

08
Geospatial
Displays data across geographic or spatial layouts e.g., population by state or floor
→ Choropleth map, cartogram

Tufte’s Principles

Edward Tufte, a pioneer in data visualization, proposed a set of principles to guide the creation of **clear**, **honest**, and **efficient** visual representations of data.

He argued that graphics should communicate data directly and efficiently, without unnecessary decoration or distraction. Instead of focusing on design flair, Tufte advocated for visuals that induce the viewer to focus on the substance of the data itself.

Tufte's Principles

Tufte states that graphical displays should:

- show the data
- induce the viewer to think about the substance rather than about methodology, graphic design, the technology of graphic production, or something else
- avoid distorting what the data has to say
- present many numbers in a small space
- make large data sets coherent
- encourage the eye to compare different pieces of data
- reveal the data at several levels of detail, from a broad overview to the fine structure
- serve a reasonably clear purpose: description, exploration, tabulation, or decoration
- be closely integrated with the statistical and verbal descriptions of a data set.

THANK YOU